Task-Conditioned Residual Skills for Reliable Multi-Stage Contact Manipulation

An Evidence-Driven Research Manuscript

ARES Research Working Group

Working manuscript, 27 October 2026; not submitted or peer reviewed

Abstract

Multi-stage contact manipulation requires more than reaching intermediate positions: object orientation, motion and tool contact persist into subsequent stages. We study constrained residual reinforcement learning on nominal robot control, using task-relative geometry and past execution effects as conditioning signals. A JAKA testbed executes two-stage free-object pushing (P1) and guided-slot insertion (P2) in 16 independent PhysX environments. Historical three-seed evaluation yields 13, 18 and 23 successes out of 32 unseen-geometry cases for ordinary residual PPO versus 8 for damped feedback, but P2 alone yields only 6, 6 and 7 out of 16 versus 5. A new matched-budget P2 development pilot compares BODY-XY residuals, task-frame residuals and effect-history conditioning using Isaac Sim 5.1 and Isaac Lab 2.3.0. With 16,384 interactions per main model, their independently accepted counts are 30/29/30, 30/30/30 and 30/30/30 out of 32; corrected task-local feedback also achieves 30/32. Input interventions change applied references and object trajectories but preserve successful-case sets. The outcome-aware increment therefore fails its preregistered promotion criterion. The new cohort accounts for 196,608 online interactions including retained software diagnostics. These results establish executable training and a useful engineering baseline, not a validated novel algorithmic advantage. Further progress requires genuinely actionable contact parameters and independent mechanism evidence. No real-robot, visual-policy or hardware-safety claim is made.

Keywords: contact manipulation; residual reinforcement learning; skill parameters; task conditioning; reproducible robotics.

Evidence status. Historical and new pilot success counts are independently audited. The new pilot is development evidence, not a final test. The outcome-history increment did not outperform its matched task-frame baseline.

1 Introduction

Industrial robot skills are often implemented as a combination of geometric planning, feedback control and task-specific execution contracts. These components can provide reliable approach and transport motions, while sustained object contact remains sensitive to object motion, orientation, friction and tool placement. A locally successful preparation motion need not leave a state from which the next operation finishes stably. Reinforcement learning (RL) offers a way to adjust executable actions using physical task feedback without requiring it to replace inverse kinematics or collision-aware planning.

Our application context is an existing industrial skill architecture, ARES/ARES-R. The present algorithmic experiments, however, are deliberately narrower than that system: a single JAKA arm manipulates a free synthetic object with a research paddle in two consecutive contact stages. They are not claims of tray grasping, production cabinet operation or end-to-end autonomous field execution. Controlled October field records are retained on a separate reference track and are excluded from this public package.

Previous experiments already show that residual PPO can learn these tasks. They also expose an important methodological risk. Aggregate performance on a mixture of tasks can hide a weak result on the more contact-sensitive task, and an auxiliary network or learned switch can update without improving physical outcomes. On unseen geometry, residual PPO improves the aggregate result over a tuned controller, but most of the gain is attributable to P1. Simple waiting at handoff and learned holding do not deliver an independent advantage. These negative findings motivate a bounded investigation rather than further scaling of a weak mechanism.

We focus on the following question: *Can task-relative legal residual parameters, informed by observed execution effects and subsequent task requirements, improve guided contact completion beyond both a strong task-frame controller and ordinary residual RL?* This does not require showing that RL is indispensable. It requires a measurable, repeatable contribution under equivalent information and interaction budgets.

The currently supported contributions are an auditable contact-RL experimental basis and a three-seed historical residual-control comparison. The proposed contribution is an execution-result-aware residual skill mechanism. Its novelty and performance remain to be established through a task-frame baseline, causal action interventions, conditioning ablations and a new held-out evaluation. Merely concatenating context, rotating coordinates or training a predictor is not considered a sufficient algorithmic contribution.

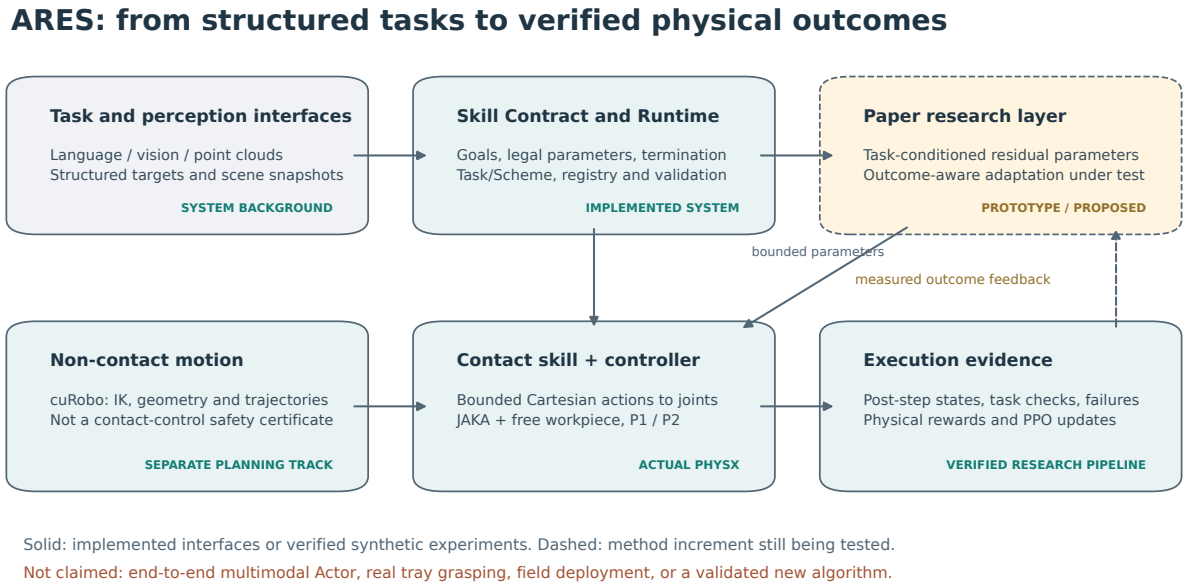


Figure 1: Scope of the study within the ARES skill architecture. The contact-RL experiment is implemented; multimodal policy deployment and long-horizon interfaces are not established results.

2 Related Work and Novelty Boundary

Parameterized actions and skill composition. RAPS studies parameterized action primitives as an RL action interface [3]. STAP uses task-agnostic skill values and planning to reason about geometric dependencies in skill sequences [1]. These precedents rule out claims that parameterized skills or successor feasibility are themselves new. The question here is narrower: whether observed sustained-contact effects can improve a constrained residual decision in a physically continuous task, beyond ordinary residual RL and task-relative control.

Residual and geometry-aware skills. Residual Skill Policies already learn residual corrections in skill space [7]. HACMan++ studies contact-rich manipulation through actionable contact/action abstractions [6]. GRN addresses feasibility-related reasoning [2], while Fabrica combines planning and geometrically structured residual skills [9]. Therefore, using a nominal controller, task frame, geometry input or feasibility mask cannot independently support novelty. A successful extension would need to show that a particular execution-feedback pathway changes actions and improves P2 under matched resources, with competing explanations removed.

Recovery and heterogeneous composition. RecoveryChaining combines nominal policies with learned recovery [10]. TASP studies the composition of heterogeneous black-box skills [4]. Our experiments do not yet implement a high-level recovery scheduler or reproduce these benchmarks. A future recovery mechanism must start from genuine recoverable physical states and be compared with legitimate retry and re-contact controllers.

Implications for this manuscript. The supplied literature comparison documents mechanism overlap, available implementation evidence and replication costs. The empirical boundary is strict: existing ARES results are self-built simulation experiments, not official benchmark scores. The task-frame change is an engineering/inductive-bias baseline. Only a reproducible increment over that baseline, linked to a genuine outcome-conditioning intervention, could justify a stronger method claim. None of the present text claims the first use of residual RL, skill chaining, successor value or learned termination.

3 Problem Formulation

We consider a finite-horizon controlled physical process with robot state $q, \dot{q}$, free-object pose x_o, R_o, object velocities v_o, ω_o, current stage k, task geometry g, and a skill contract C. The contract describes permissible action dimensions, reference-motion bounds, workspace, orientation requirements and task termination. The simulator advances at 120 Hz and control actions at 60 Hz. Stage transitions preserve robot, object and contact dynamics; they do not reset the object to an advantageous state.

Observation and information. The historical actor receives a 56-dimensional state-based observation including robot state, object pose and velocities, object–tool relation, stage and subsequent target errors, a simulated net-contact proxy and known geometry/dynamics context. Exact friction and mass context are assumed

available in simulation. This is not a learned visual observation or an unknown-parameter identification result. Future suffix success and independent acceptance labels are never actor inputs. A proposed feedback history h_t may contain only already observed execution effects.

Task-relative action. Let T_g express the local task frame and ξ_{nom} a nominal contact reference increment. A residual parameter policy yields

$$\xi_t = \mathcal{P}_{C_t}(\xi_{\text{nom}}(o_t, C_t) + f_\theta(o_t, C_t, C_{t+1}, h_t)), \quad (1)$$

where $\mathcal{P}_{C_t}$ applies executable bounds to the *combined* command. A DLS controller maps the reference to constrained joint commands. Joint soft limits, tool orientation and reference lead bounds remain active for both nominal and learning components. Task-frame conversion alone is baseline B2, not the proposed innovation.

Objective and endpoint. The primary endpoint is independently accepted complete P2 success, not reward or intermediate position alone. The v6 contract requires target error below 8 mm, speed below 10 mm/s, angular speed below 0.2 rad/s, square-object symmetry yaw error below 8°, upright error below 5°, object height within 8 mm of the supported reference height, and 0.5 s of continuous stable completion, within a 12 s deadline. P1 is reported separately as a foundational task. A sampled geometric/process audit complements the environment's success flag.

The historical reward was fixed before its formal comparison:

$$\begin{aligned} r_t = {} & 60\Delta d_t - \Delta t - 0.0002\|a_t\|^2 \\ & + 3I_{\text{switch}} + 10I_{\text{success}} - 5I_{\text{violation}}. \end{aligned} \quad (2)$$

Progress is recomputed within each stage without counting a reset target distance as instantaneous achieved progress. This stagewise shaping is *not* claimed to be a globally policy-invariant potential difference: a phase-dependent potential would require an explicit jump correction. The new pilot uses the same stagewise progress and terminal/time terms across all learned controls, with action cost based on the applied reference rather than an unexecuted proposal. Reward changes are not an algorithmic contribution. Potential-based shaping $\gamma\Phi(s_{t+1}) - \Phi(s_t)$ remains an untested alternative. Current contact policies use fixed control intervals and ordinary MDP PPO; unequal-duration high-level options would require explicit SMDP discounting and are not asserted here.

4 Implemented Basis and Candidate Method

4.1 Nominal control and ordinary residual PPO

The damped nominal controller tracks a stage target while respecting a fixed legitimate pushing side. Reference lead over the actual tool is bounded to 6 mm to prevent integral windup. Ordinary residual PPO adds at most 0.6 mm per control step in planar reference corrections. All methods share the same DLS joint controller and physical constraints. This strong nominal policy is retained and improved alongside any task-frame transformation; it is not weakened to manufacture an RL advantage.

The historical learning comparison uses 16 environments, a [64, 64] MLP, rollout length 256, six minibatch epochs and $\gamma = 1$. Each main model receives 65,536 online interactions. Final checkpoints are fixed by interaction budget, not selected on the frozen tests. PPO optimizes the usual clipped surrogate with GAE. A separate diagnostic mixes GAE with a current-rollout, value-bootstrapped $\lambda = 1$ target; it does not reuse stale episodes across policy updates and is not presented as a new credit-assignment method.

For clarity, the policy update is ordinary PPO [8], with likelihood ratio $r_t(\theta) = \pi_\theta(a_t|o_t)/\pi_{\text{old}}(a_t|o_t)$:

$$L_{\text{actor}} = \mathbb{E}_t\left[\min\{r_t\hat{A}_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\hat{A}_t\}\right]. \quad (3)$$

Context or history changes this distribution's inputs, not the estimator into a new credit-assignment algorithm. The value loss and optimizer use the same data budget as their matched baselines.

4.2 Task-frame baseline B2

The new development comparison converts task errors, velocities, object orientation and permissible reference directions to the local entrance frame. B1 retrains the BODY-XY residual on the same new physical cases; B0 is the repaired task-local controller. B1/B2 share source physical state, input size, capacity, physical action range, reward and interaction budget, but exact information equivalence is not guaranteed. In particular, B2 canonicalizes the explicit entrance-direction cosine/sine to 1/0 while joint state and physical workspace remain in BODY coordinates. This is neither a strictly invertible whole-state transform nor a demonstrated rigidly equivariant policy. Any B1/B2 difference is a bundled engineering/representation comparison, not an isolated proof of frame causality. B2 and B3 retain the same base representation, supporting the targeted history ablation.

4.3 Outcome-aware candidate B3

The implemented B3 pilot adds eight one-step history channels to the historical 56-state basis: task-forward/lateral object finite-difference velocity (two), previous applied reference increment divided by control duration (two), their differences (two), object angular velocity about the vertical axis (one), and the previous normalized phase-0 contact-target command (one). The last quantity is a requested parameter, not an observed workpiece displacement. Each channel is bounded in observation preprocessing. These signals directly enter the actor; no future success label, learned predictor or disconnected auxiliary loss is used.

All learned pilot methods have three continuous outputs: planar residual correction bounded to 0.6

mm/control step and a phase-0 lateral contact-target offset bounded to 4 mm. Only the tool reference changes, not the physical object or task goal. B1/B2 set the extra eight history channels to zero but retain the same 64-dimensional input and network. They still observe current pose, velocity and contact proxy, so this ablation is called *no effect-history*, not absence of all physical outcome information. The downstream-goal ablation masks phase-0 future-target and some entrance-geometry channels while retaining mass/friction context; the legal action mapping still uses the known task direction. It is a *partial* next-contract intervention, not a policy with no downstream information.

Writing R_g for the planar task rotation, the measurable response part is formed as $R_g^\top (x_{o,t} - x_{o,t-1})/\Delta t$ and $R_g^\top \Delta x_{\mathrm{ref},t-1}/\Delta t$, together with their difference. The requested contact parameter is $4\,\mathrm{mm}\ \mathrm{clip}(a_{t,3}, -1, 1)$ in phase 0. It changes the controller's lateral contact target while the physical workpiece goal remains unchanged; rear-face alignment and reference antiwindup can suppress its instantaneous physical effect. A nonzero parameter command alone is not executed displacement. A cold-start paired probe records actual applied references and object trajectories to test whether it affects subsequent results.

This is ordinary full-chain-reward PPO with a real parameter-action interface. It does not implement a learned successor Q-function or learned stage termination, and history concatenation alone is not a new algorithm. Its scientific status is a prototype whose independent benefit must be tested. The actual run-source snapshots and protocol define the implementation; a single identified pre-fix observation-index error is preserved as diagnostic data and excluded from the corrected formal pilot.

4.4 Falsifiable ablations

The mechanism comparison removes outcome history, removes the next contract, provides capacity-matched ineffective conditioning, or trains a predictor without connecting it to policy decisions. These controls are required to separate information, representation, extra data and actual mechanism effects. Not-yet-run cells remain explicitly untested. A finite search or supervised scorer is included only when the implemented parameter decision has a well-defined candidate interface; a one-shot choice is labelled a contextual bandit rather than long-horizon RL.

5 Experimental Protocol

5.1 Physical tasks and evidence levels

P1 is two-stage free-cube pushing. P2 uses a 40 mm cube, a separate 30 mm research paddle and 100 mm guide rails, with a target sufficiently deep for the cube to enter the guide. It is not precision insertion of a measured industrial tray. Mass, inertia, friction, entrance direction, initial yaw and target vary through actual PhysX properties or geometry. Original robot/gripper collisions remain; the paddle's merged tool inertia retains a historical approximation.

One Isaac Lab 2.3.0 application [5] hosts independent cloned environments rather than 16 separate applications. Actions change robot references and joint drives through real physics integration; object states are not rewritten during action execution. The experiment is synthetic/state-based/actual PhysX. The ARES-R field-reference track remains separate and is neither a training-label source nor an openly published scene.

5.2 Splits, budgets and evaluation

Historical v6 uses eight train groups, two validation groups and distinct ID, geometry and dynamics test groups. Three seeds (31, 41, 51) are evaluated on the same 32 cases per split. These historical tests have been read and cannot be reused for selecting new architecture or thresholds. The new pilot uses newly declared development groups, followed by a separately sealed final set only if the mechanism warrants promotion.

B0 is tuned legal damped/task-frame feedback; B1 is BODY-XY residual PPO; B2 is task-frame residual PPO; B3 is the actual effect-history candidate; B4 is finite search or scoring only when the interface supports it. The corrected pilot fixes seeds 31/41/51 and 16,384 online interactions per model, followed by 32 new P2 validation cases with case seed 102741. All learned methods use the same 64-dimensional input shape, two 64-unit hidden layers and three Gaussian outputs. A failed pilot is retained, not replaced by a favorable seed. P2-only completion is the primary metric; P1, time, contact/rotation diagnostics, interaction count and planning budget are secondary.

The preregistered promotion criterion requires a positive B3–B2 P2 count difference in all three paired seeds, mean improvement of at least two cases out of 32, and conditioning interventions with a consistent behavioral/result contribution. Otherwise the mechanism remains a prototype and is not expanded to five seeds or final-test selection. A fixed $-4/0/+4$ mm phase-0 contact-parameter experiment performs three complete physical executions per case; its hindsight union is not an equal-budget online search baseline.

5.3 Independent process checks and statistics

Exact post-step terminal state is saved before reset with physics ticks. A 120 Hz sampled FCL/OBB check covers tool, cube, table and rails, plus FK/soft-limit and final stability checks. A separate mesh diagnostic samples actual robot collision geometry in a limited subset. Neither is continuous-collision detection or a hardware safety certificate. A simulated 80 N net-contact bound is uncalibrated for hardware.

We report per-seed counts and paired case differences. Seed sample standard deviation is not a binomial confidence interval, and repeated evaluation of the same

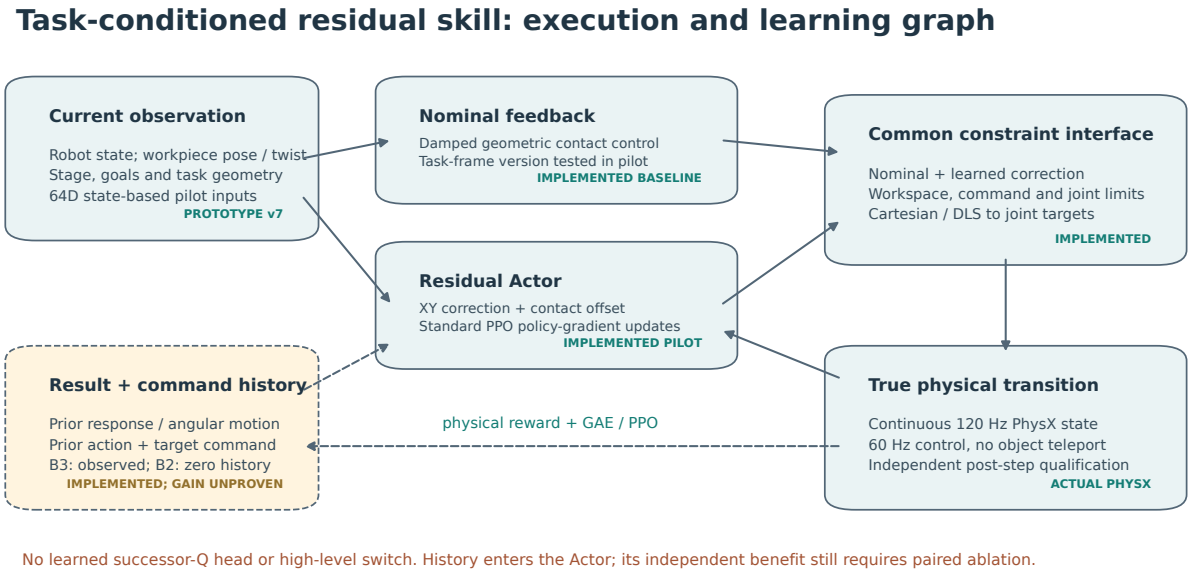


Figure 2: Computation and evidence flow. Nominal control, constraints, state-based PPO and physical feedback are implemented. The outcome-history prototype enters the actor, but a performance contribution requires matched physical ablations; a disconnected auxiliary head is not a contribution.

Table 1: Historical environment and evidence scope. The new pilot extends the actor to 64 inputs while retaining common physical state and constraint interfaces. Generated source: Tab1_environment.tex.

Component	Specification	Evidence boundary
Physics / control	Isaac Sim 5.1; Isaac Lab 2.3.0; 120 / 60 Hz	Verified synthetic simulation
P1 / P2	Free-workpiece two-stage push / deep guided slot	Self-built; not official benchmark
Robot / object	Single JAKA; 40 mm cube; 30 mm research paddle	No real tray grasping claim
Actor / policy	56D state-based observation; MLP 64–64	Not vision deployment
Task acceptance	Post-step state and 30 controls stable completion	Independent task/geometry checks
Formal cohort	3 methods x 3 seeds x 65,536 interactions	Historical formal training
Full historical cohort	901,120 interactions; 16 trained runs including diagnostics	Not 16 formal seeds

cases under multiple models does not create independent environments. Time comparisons use explicitly identified common-success cases. Additional prediction/suffix interaction, demonstration continuation and diagnostic runs are accounted for separately. Software-incomplete runs are not assigned fabricated physical failure labels.

6 Results and Research Decisions

6.1 Verified historical starting point

Table 2 reports independently accepted completions, not raw position hits. Ordinary residual PPO improves unseen-geometry aggregate completion from feedback's 25% to a seed mean of 56.25%, with sample standard deviation 15.63 percentage points. This is preliminary residual adaptation evidence, not a novel-method result. ID and dynamics results are close to feedback.

The more informative P2 breakdown is much less favorable: residual PPO obtains 6/6/7 successes out of 16, compared with feedback's 5. P1 accounts for a large share of aggregate improvement. Across common geometry successes, residual completion is 1.10–1.31 s faster depending on seed, but this is conditioned on joint success, not an unconditional task-time advantage. All models cover both tasks; a shared-training benefit has not been measured against single-task training.

6.2 Negative mechanism evidence

Paired complete physical replays compare waiting 0, 12 and 30 control steps in a legitimate handoff region before an identical suffix controller. Among 46 common valid handoffs, suffix successes are 29, 27 and 27 while average handoff speed decreases from 28.7 to 21.2 and 7.1 mm/s.

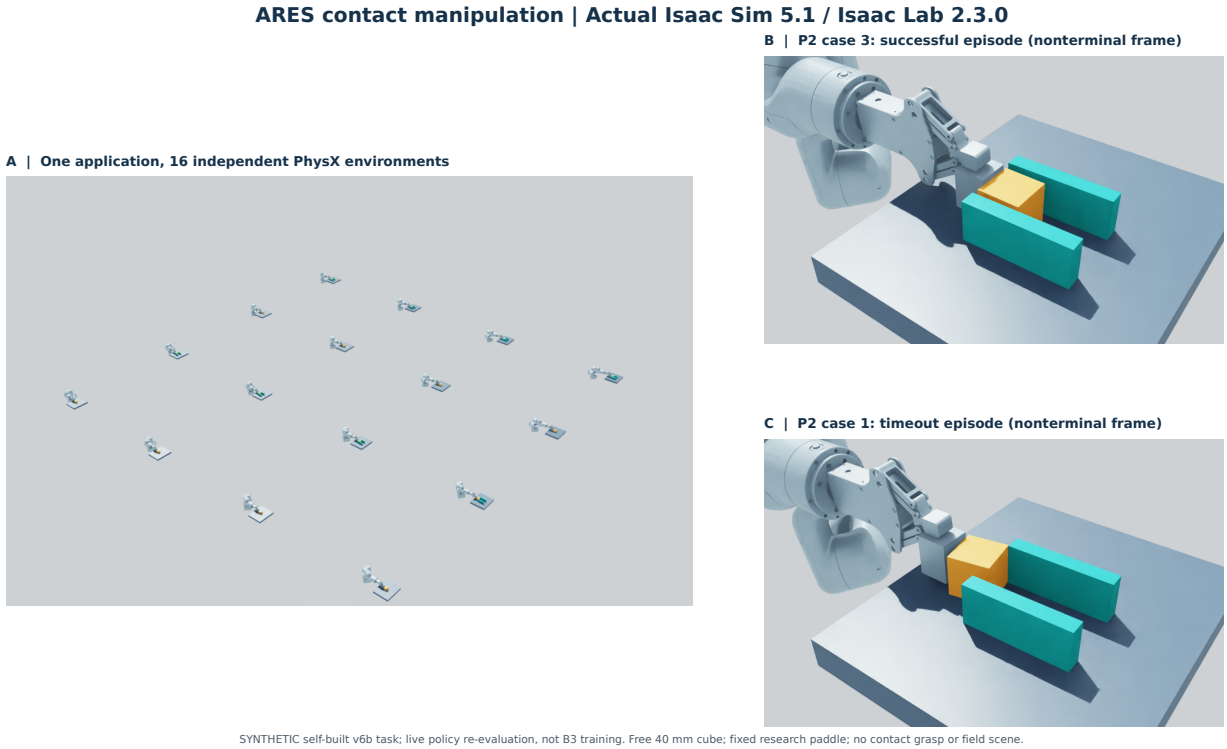


Figure 3: Actual Isaac Sim 5.1/Isaac Lab live physics evaluation of the historical v6b residual seed31 policy, not new B3 training or recorded-trajectory playback. Case3 is shown at physics tick800 (6.67 s) and later passes independent acceptance; case1 is shown at tick1184 (9.87 s) and later times out. Both images are nonterminal frames. The overview contains 16 independent environments. Physical attribute signatures are unchanged by render styling. The complete live evaluation has 55/64 independently accepted successes.

Waiting alters several state variables, so velocity is not isolated causally; the experiment does not support a general waiting benefit.

Learned hold/proceed is not consistently better than ordinary residual PPO. Forcing its seed31 policy to resume preserves both the 54/64 validation count and successful-case set. A separate active-likelihood hybrid pilot changes the actual actor surrogate, but yields only 56 versus 57/64; it fails the predeclared four-case improvement threshold. These outcomes are retained as negative evidence rather than described as successful innovations.

6.3 New development pilot

The new cohort is a development split and cannot be pooled with historical tests. Each B1/B2/B3 model receives 16,384 online interactions. Independent P2 success on 32 matched validation cases is 30/29/30 for B1, 30/30/30 for B2 and 30/30/30 for B3; corrected B0 feedback achieves 30/32. The paired B3–B2 count differences are 0/0/0. Thus the preregistered requirement of a positive difference in every seed and a mean gain of at least two cases is not met. No five-seed expansion or final-test model selection is performed. The much higher new P2 completion cannot be attributed to a novel algorithm: this cohort uses different geometry groups and a repaired shared nominal controller.

For seed31, removing all eight history channels, removing measured response while retaining command memory, or partially masking next-contract inputs preserves the same 30/32 successful-case set. Removing history changes the mean per-case applied-reference-increment RMS by about 20.6 micrometres and object-trajectory RMS by 54.2 micrometres. All compared initial robot/object/tool states match exactly in the recorded reset variables. These are real action/trajectory effects, not task-completion benefits; subsequent states are not forced to match and PhysX contact caches are not claimed to be restored. A separately trained no-next seed31 model also achieves 30/32. Fixed contact-target probes do not establish a recovery advantage and cost three full executions per case rather than one.

The new budget is 147,456 interactions for nine main models plus 16,384 for a separately trained ablation (163,840 formal), and 32,768 for two retained pre-fix software diagnostics: 196,608 new online interactions and 12 trained models in total. They are not the historical 901,120 interactions. There are 624 completed formal evaluation episodes, including training-geometry checks. Actor optimizer records show actual nonzero updates; physical trajectory and task checks are separate evidence. Persistent failures occur during phase1, approximately 29–38 mm from the goal with simulated net-contact proxies of 45–59 N, rather than only failing the final stability window.

The $-4/+4$ mm contact-target probes change mean handoff object position by approximately 1.3–1.5 mm relative to zero offset; they finish 30/29 cases versus 30 at zero. The positive offset loses one case and nei-

Table 2: Historical v6 independently accepted success generated from audited case records. Each split has 32 matched cases, comprising 16 P1 and 16 P2 cases; triples are seeds 31/41/51, not independent case replications. P1 cannot substitute for P2 evidence. These are not new pilot results.

Method	Task	ID	Geometry	Dynamics
Damped feedback	All	29 / 32	8 / 32	28 / 32
	P1	16 / 16	3 / 16	16 / 16
	P2	13 / 16	5 / 16	12 / 16
Direct PPO	All	0, 0, 0 / 32	0, 0, 0 / 32	0, 0, 0 / 32
	P1	0, 0, 0 / 16	0, 0, 0 / 16	0, 0, 0 / 16
	P2	0, 0, 0 / 16	0, 0, 0 / 16	0, 0, 0 / 16
Residual PPO	All	28, 28, 29 / 32	13, 18, 23 / 32	28, 29, 29 / 32
	P1	16, 16, 16 / 16	7, 12, 16 / 16	16, 16, 16 / 16
	P2	12, 12, 13 / 16	6, 6, 7 / 16	12, 13, 13 / 16
Residual + hold/proceed	All	28, 30, 30 / 32	15, 15, 18 / 32	28, 29, 29 / 32
	P1	16, 16, 16 / 16	9, 8, 12 / 16	16, 16, 16 / 16
	P2	12, 14, 14 / 16	6, 7, 6 / 16	12, 13, 13 / 16

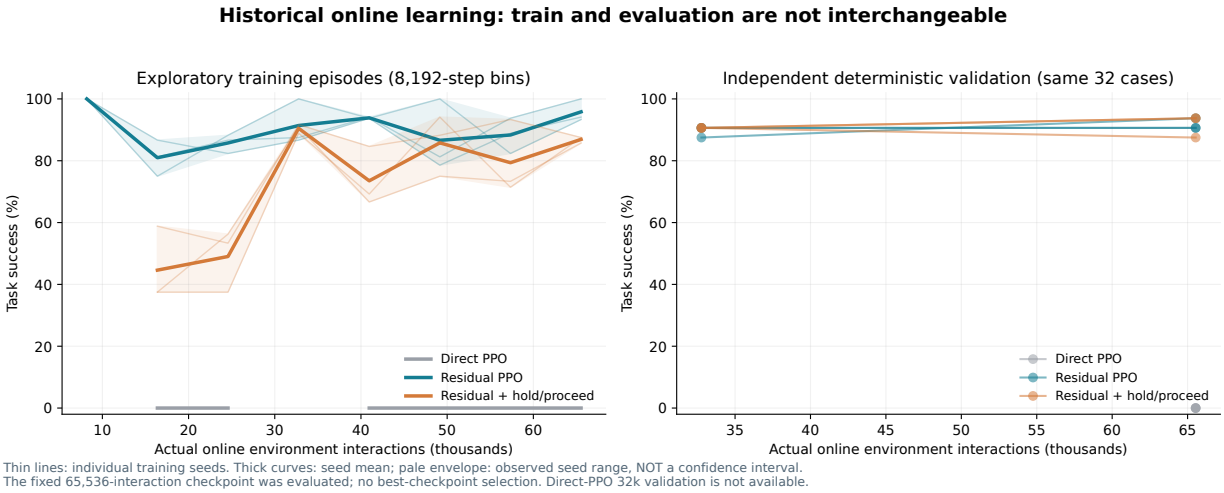


Figure 4: Actual historical online training curves. Seed traces and interaction counts are shown; training reward and frozen deterministic task success are distinct measures.

ther probe rescues a failure. Thus a nonzero command has some physical authority but is not a measured 4 mm displacement or a useful outcome mechanism here. A bounded existing withdrawal rule triggers three attempts on each of the two persistent failures and rescues neither; this does not prove geometric impossibility or absence of other recoveries.

Common-success mean times for B2 versus B3 are 9.3022 versus 9.1333 s, 9.4650 versus 9.2161 s, and 9.3711 versus 9.3217 s for the three seeds. The nominal controller takes 9.2461 s and the same B3 checkpoint without history takes 9.1294 s. The small B3–B2 timing difference therefore does not establish an effect-history mechanism or consistent superiority to strong feedback, and it cannot replace the preregistered primary-success criterion.

6.4 Training correctness

The old episode-only credit prototype modified just 23.39% of seed31 samples; its boundary and all-stage modes coincided because eligible episodes ended in stage0. Corrected current-rollout bootstrapping has complete coverage but is not a stage-specific innovation. Exact post-step logging removes a terminal velocity timing ambiguity without relaxing thresholds. Final deterministic versus stochastic replay of an old model yields 51 versus 28/64, illustrating why early exploratory training episodes cannot be compared directly with a final mean policy. Direct PPO remains unreliable even after removing an unused action dimension; its zero final result is not evidence that all reasonable direct RL methods fail.

7 Limitations and Conclusion

This study establishes a reproducible physical contact-RL basis and a limited historical residual-control geometry result. It does not establish a novel outcome-aware algorithm: the new effect-history pilot fails its promotion criterion against matched task-frame residual PPO. Each pilot model has only four rollout updates and 16 completed training episodes; the bounded null result does not disprove execution-history RL in general. The

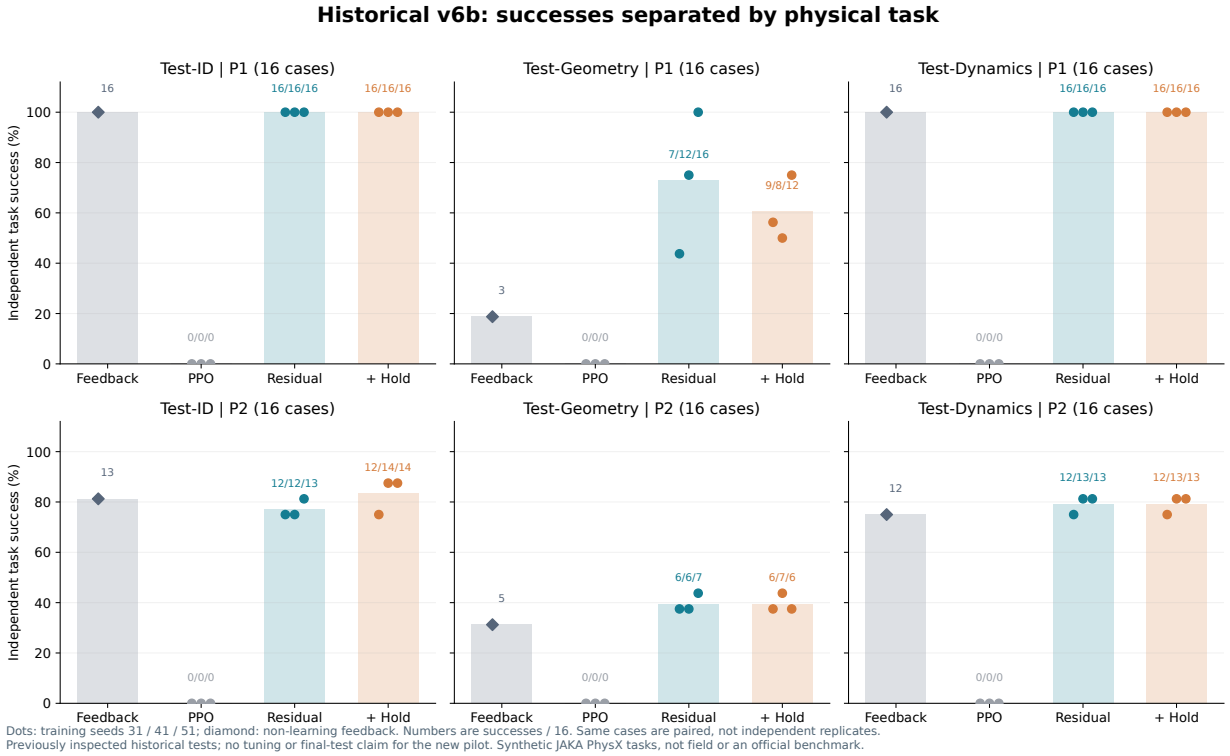


Figure 5: Historical frozen physical evaluations with per-seed and P1/P2 decomposition. The original tests are now descriptive evidence, not a new method-development validation set.

historical three-seed geometry result has substantial variance and limited P2 improvement. The new near-ceiling development set does not establish unseen-condition generalization, while direct PPO needs further matched tuning and rotation/shared-training ablations remain untested. Historical aggregate improvement must not be represented as a solved narrow-slot task.

State-based simulator observations include privileged pose and known dynamics context. A deployed perception system, visual/point-cloud actor, measured tool dynamics, true grasping, full robot CCD and real-robot validation have not been demonstrated. The synthetic cube and research paddle are not the field tray. Field references are kept private and separate. Long-horizon option interfaces, VLA instructions and skill recovery are future extensions, not completed contributions.

The next decision is deliberately bounded: do not add history channels or extend training on the near-ceiling pilot. First establish physical authority of a re-contact/contact-location parameter under cold-start paired executions, then compare the same strong feedback and residual controls in physically justified, non-saturated contact conditions. Promote only a repeatable P2 gain with a valid mechanism ablation. Negative results narrow this mechanism without discarding the platform or the requirement to study RL. A publishable application-method paper still requires an independently supported increment, more seeds, sealed evaluation groups and stronger related-method comparisons.

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Table 3: New P2 development pilot from actual run records. Reported successes are independently accepted. Interactions denotes the underlying policy's training budget: inference interventions do not incur that training budget again. Nonlearning probes have zero training, not zero physical evaluation cost. These cases are not a final test and must not be pooled with historical tests.

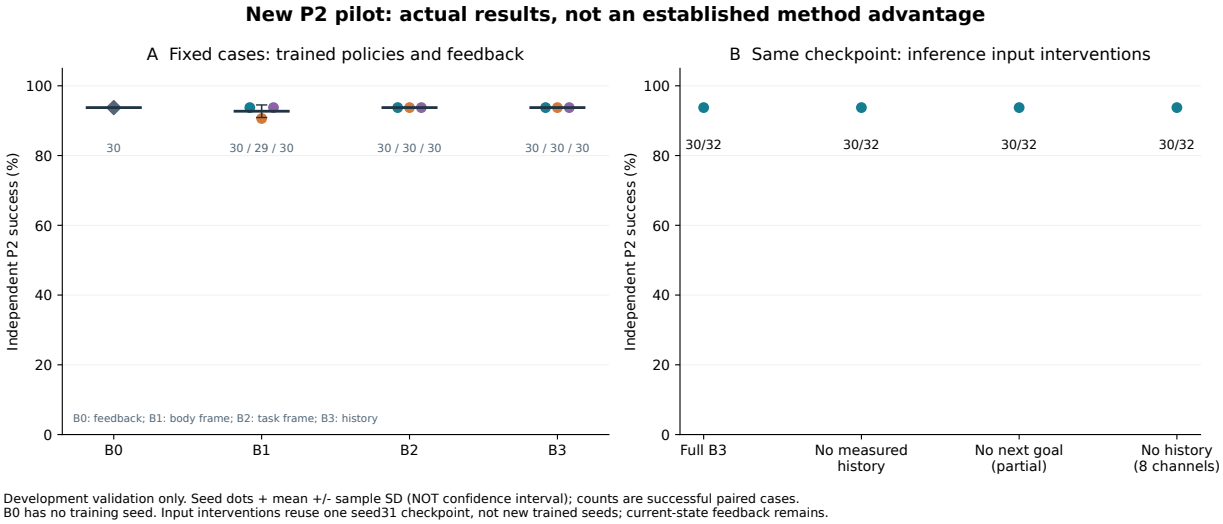
Method	Seed	Task	Independent success	Model train steps
B0	31	P2	30/32	0
Feedback + backoff	31	P2	30/32	0
B1	31	P2	30/32	16384
B1	41	P2	29/32	16384
B1	51	P2	30/32	16384
B2	31	P2	30/32	16384
B2	41	P2	30/32	16384
B2	51	P2	30/32	16384
B3 measured-history mask (eval)	31	P2	30/32	16384
B3 next-goal mask (eval)	31	P2	30/32	16384
B3 no next goal (train)	31	P2	30/32	16384
B3 history mask (eval)	31	P2	30/32	16384
B3	31	P2	30/32	16384
B3	41	P2	30/32	16384
B3	51	P2	30/32	16384
Fixed contact +4 mm	31	P2	29/32	0
Fixed contact -4 mm	31	P2	30/32	0

Table 4: Mechanism interventions and resource accounting generated from the actual pilot evidence. An input or action effect is not by itself evidence of improved completion.

Variant	Frame	Result/command history	Next goal	Seeds	Train budget
B0	Task	Zero / none	Observed	31	0
Feedback + backoff	Task	Zero / none	Observed	31	0
B1	Body	Zero / none	Observed	31,41,51	16384
B2	Task	Zero / none	Observed	31,41,51	16384
B3	Task	Observed	Observed	31,41,51	16384
B3 no measured effect (eval)	Task	Command memory only	Observed	31	Same checkpoint
B3 no next goal (train)	Task	Observed	Hidden during training	31	16384
B3 no next goal (eval)	Task	Observed	Suppressed at eval	31	Same checkpoint
B3 no history (eval)	Task	Suppressed at eval	Observed	31	Same checkpoint
Fixed contact +4 mm	Task	Zero / none	Observed	31	0
Fixed contact -4 mm	Task	Zero / none	Observed	31	0

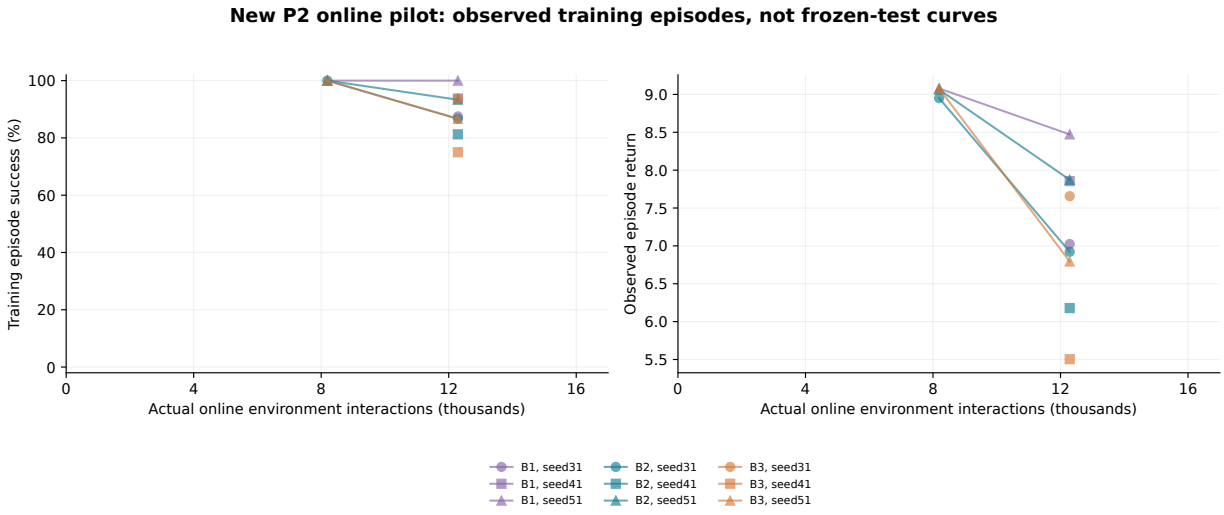
Javid, Branden Romero, Sachin Chitta, Shinjiro Sueda, Hui Li, and Wojciech Matusik. Fabrica: Dual-arm assembly of general multi-part objects via integrated planning and learning. In *Proceedings of the 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 406–419, 2025.

[10] Shivam Vats, Devesh K. Jha, Maxim Likhachev, Oliver Kroemer, and Diego Romeres. RecoveryChaining: Learning local recovery policies for robust manipulation. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025.



Development validation only. Seed dots + mean +/- sample SD (NOT confidence interval); counts are successful paired cases.
B0 has no training seed. Input interventions reuse one seed31 checkpoint, not new trained seeds; current-state feedback remains.

Figure 6: Actual new P2 development evaluation and same-checkpoint conditioning interventions. B3 does not improve over B2. Mask interventions are evaluations of one trained model, not additional training seeds. Independent physical outcomes, not expected gains, populate this figure.



Completed episodes binned by 4,096 actual interactions; empty bins remain gaps. Lines are individual seeds, not confidence intervals.
A short pilot and high nominal-control success do not establish convergence or a method advantage.

Figure 7: Actual new online training episode curves under the fixed pilot budget. Exploratory episode reward/success is distinct from final deterministic independent validation. All extra diagnostic and ablation interaction budgets are retained.